

Exercise 12.1

Consider the following controllable and observable system

$$\begin{cases} \dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 1 \end{bmatrix} x. \end{cases}$$

and associated controller and observer with gains

$$F = \begin{bmatrix} -1 & -1 \end{bmatrix} \quad L = \begin{bmatrix} 2 \\ -9 \end{bmatrix}.$$

1. Add a reference to the system and generate a Bode plot of it. The system is given by

$$\begin{cases} \begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & BF \\ -LC & A + BF + LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} r \\ y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}. \end{cases}$$

2. Where is it appropriate to place zeros?
3. Place zeros, by designing the matrix $\tilde{M} = MN^{-1}$.
4. Compute the matrix N to obtain the desired DC-gain.
5. Compute $M = \tilde{M}N$.
6. Generate a Bode plot for the designed system, and compare it to the original one. Recall that the system equations are

$$\begin{cases} \begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & BF \\ -LC & A + BF + LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} BN \\ M \end{bmatrix} r \\ y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}. \end{cases}$$